

LECTURE 4 · WEB TECHNOLOGIES

Introduction to CSS

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What is CSS?

- CSS stands for Cascading Style Sheets
- Styles define how to display HTML elements
- Styles were added to HTML 4.0 to solve a problem
- External Style Sheets can save a lot of work
- External Style Sheets are stored in CSS files
- Styles Solved a Big Problem

What is CSS?

- CSS removes the presentation attributes from the structure allowing reusability, ease of maintainability, and an interchangeable presentation layer.
- **HTML was never meant to be a presentation language.** Proprietary vendors have created tags to add presentation to structure.

  <font>

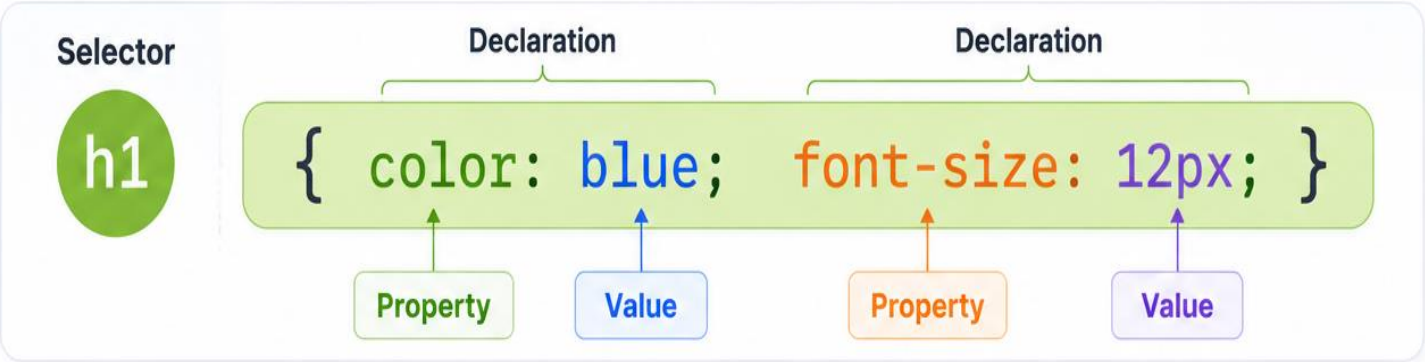
  <b>

  <i>
- CSS allows us to make global and instantaneous changes easily.

CSS Syntax

- selector/element {
  property: value;

}



- Example

p {color:red;text-align:center;}

CSS Comments

- Comments are used to explain your code, and may help you when you edit the source code at a later date. Comments are ignored by browsers.
- A CSS comment begins with "/*", and ends with "*/", like this:

```
p
{
text-align:center;
/*This is another comment*/
color:black;
font-family:arial;
}
```

CSS Selectors

- Selectors are one of the most important aspects of CSS as they allow you to target specific elements on your web page in various ways so that they can be styled.

- The selector can
  - an identifier,(id)
  - class
  - single XHTML element (body, div, etc)
  - a grouping of elements

Single HTML element

- Specify the style(s) for a single HTML element.

```
body
{
 margin: 0;
 padding: 0;
 border-top: 1px solid #ff0;
}
```

Grouping of elements

- Allows you to specify a single style for multiple elements at one time.

```
h1, h2, h3, h4, h5, h6
{
  font-family: “Trebuchet MS”, sans-serif;
}
```

Class Selector

- The class selector is used to specify a style for a group of elements. Unlike the id selector, the class selector is most often used on several elements.
- This allows you to set a particular style for any HTML elements with the same class.
- The class selector uses the HTML class attribute, and is defined with a ".“
- Example

```
<p class=“intro”>This is my introduction text</p>

.intro {
  font: 12px verdana, sans-serif;
  margin: 10px;
}
```

Id Selector

- The id selector is used to specify a style for a single, unique element.
- The id selector uses the id attribute of the HTML element, and is defined with a "#"
- Example

```
<p id=“intro”> This is my introduction text</p>

#intro {
   border-bottom: 2px dashed #fff;
}
```

Selectors

- Identifier or class? What are the differences?
  - An identifier is specified only once on a page and has a higher inheritance specificity than a class.
  - A class is reusable as many times as needed in a page.
  - Use identifiers for main sections and class for sub-sections of your document

Advanced CSS Selectors

- Descendant Selector
  body h1 { } - - Selects all <h1> elements inside <body> element
  #navigation p {} - # is used for id
- Child Selectors element
  div ol > p {} - Selects all <p> elements where the parent is a <ol> which is inside element
- Universal Selector
  * {} – select all elements
- Attribute Selectors
  a[href="http://home.org"]
- Class Selectors
  .intro - Selects all elements with class="intro"
- Pseudo Class Selcetor

a:hover{}
#nav:hover {}

Using Style Sheets

- External Style Sheet
  <link rel="stylesheet" type="text/css" href="location.css" />
- Embedded/Internal Styles
  <style type="text/css">
  </style>
- Inline Styles
  * Lorem ipsum *

- *multiple external style sheets can be referenced inside a single HTML document*

External Style Sheet Example

- **CSS Code**

```
body{ background-color: gray;}
p { color: blue; }
h3{ color: white; }
```

- **HTML Code**

```
<html>
<head>
<link rel="stylesheet" type="text/css" href="test.css" />
</head>
<body>
<h3> A White Header </h3>
<p> This paragraph has a blue font.
The background color of this page is gray because we changed it with CSS! </p>
</body>
</html>
```

Embedded/Internal Styles Example

- **CSS Code**

```
<html>
<head>
<style type="text/css">
        p {color: white; }
        body {background-color: black; }
</style>
</head>
<body>
    <p>White text on a black background!</p>
</body>
</html>
```

Inline Styles Example

- **CSS Code**

```
<html>
<head>
</head>
<body>
   <p style="background: blue; color: white;">A new background
   and font color with inline CSS</p>
</body>
</html>
```

Common Inline CSS Mistakes

- When using CSS inline, you must be sure not to use quotations within your inline CSS. If you use quotations the browser will interpret this as the end of your style value.
- **CSS Code**

```
<p style="background: url("yellow_rock.gif");">
This is broken</p>
<p style="background: url(yellow_rock.gif);">
This is working'</p>
```

CSS Background

- The background of your website is very important.
- CSS background properties are used to define the background effects of an element.
- CSS properties used for background effects:
  - background-color
  - background-image
  - background-repeat
  - background-attachment
  - background-position

Background Color

- Specifies the background color of an element

```
<html>
<head>
<style type="text/css">
        body  { background-color:#b0c4de; }
</style>
</head>
<body>
        <h1>My CSS web page!</h1>
        <p>Hello world! This is a W3Schools.com example.</p>
</body>
</html>
```

Background Color

- The background color can be specified by –
  - name - a color name, like "red"
  - RGB - an RGB value, like "rgb(255,0,0)"
  - Hex - a hex value, like "#ff0000“

- **CSS Code**
  - h4 { background-color: white; }
  - p { background-color: #1078E1; }
  - ul { background-color: rgb(149, 206, 145); }

Background Image

- Specifies an image to use as the background of an element.
- By default, the image is repeated so it covers the entire element.

```
<html>
<head>
<style type="text/css">
body {background-image:url('abc.gif');}
</style>
</head>
<body>
<h1>Hello World!</h1>
</body>
</html>
```

background-repeat

- By default, the background-image property repeats an image both horizontally and vertically.
- Some images should be repeated only horizontally or vertically, or they will look strange
- Possible values are
  - repeat-x : repeated only horizontally
  - repeat-y : repeated only vertically
  - no-repeat : not repeated horizontally or vertically

Background - Shorthand property

- It is also possible to specify all the properties in one single property. This is called a shorthand property.
- The shorthand property for background is simply "background":

```
body {background:#ffffff url('img_tree.png') no-repeat right top;}
```

- order of the property values are:
  - background-color
  - background-image
  - background-repeat
  - background-attachment
  - background-position

Summary

Property	Values
background-attachment	Scroll/ fixed/ inherit
background-color	color-rgb/ color-hex/ color-name/ transparent/ inherit
background-image	url(URL)/ none/ inherit
background-position	left top/ left center/ left bottom/ right top/ right center/ right bottom/ center top/ center center/center bottom/ x% y%/ xpos ypos/inherit
background-repeat	repeat/repeat-x/repeat-y/no-repeat/inherit

CSS Borders

- The CSS border properties allow you to specify the style, width, and color of an element's border.
- The **border-style** property specifies what kind of border to display.
- The following values are allowed:
  - **dotted -** Defines a dotted border
  - **dashed -** Defines a dashed border
  - **solid -** Defines a solid border
  - **double -** Defines a double border
  - **groove -** Defines a 3D grooved border. The effect depends on the border-color value
  - **ridge -** Defines a 3D ridged border. The effect depends on the border-color value
  - **inset -** Defines a 3D inset border. The effect depends on the border-color value
  - **outset -** Defines a 3D outset border. The effect depends on the border-color value
  - **none -** Defines no border
  - **hidden -** Defines a hidden border

CSS Borders

- The **border-width** property specifies the width of the four borders.
- The **border-color** property is used to set the color of the four borders.
- In CSS, there are also properties for specifying each of the borders (top, right, bottom, and left):
  - **border-top-style**
  - **border-right-style**
  - **border-bottom-style**
  - **border-left-style**

CSS Border - Shorthand Property

- To shorten the code, it is also possible to specify all the individual border properties in one property.
- The **border** property is a shorthand property for the following individual border properties:
  - border-width
  - border-style (required)
  - border-color
- Example

```
p {
  border: 5px solid red;
}
```

CSS Margins

- The CSS margin properties are used to create space around elements, outside of any defined borders.
- Margins define the distance between an element's border and the surrounding elements.
- With CSS, you have full control over the margins. CSS has properties for setting the margin for each individual side of an element (top, right, bottom, and left), and a shorthand property for setting all the margin properties in one declaration.

CSS Margins

- CSS has properties for specifying the margin for each side of an element:
  - **margin-top -** sets the top margin of an element
  - **margin-right -** sets the right margin of an element
  - **margin-bottom -** sets the bottom margin of an element
  - **margin-left -** sets the left margin of an element
- All the margin properties can have the following values:
  - **auto -** the browser calculates the margin
  - **length -** specifies a margin in px, pt, cm, etc.
  - **% -** specifies a margin in % of the width of the containing element
  - **inherit -** specifies that the margin should be inherited from the parent element

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  - **% -** specifies a margin in % of the width of the containing element
  - **inherit -** specifies that the margin should be inherited from the parent element

Margin - Shorthand Property

- The **margin** property is a shorthand property for the following individual margin properties:
  - margin-top
  - margin-right
  - margin-bottom
  - margin-left

Here is how it works:

- Example

```
p {
  margin: 25px 50px 75px 100px;
}
```

The auto Value

- You can set the margin property to auto to horizontally center the element within its container.
- The element will then take up the specified width, and the remaining space will be split equally between the left and right margins.
- Example:

```
div {
 width: 300px;
 margin: auto;
 border: 1px solid red;
}
```

CSS Padding

- CSS has properties for specifying the padding for each side of an element:
  - **padding-top -** sets the top padding of an element
  - **padding-right -** sets the right padding of an element
  - **padding-bottom -** sets the bottom padding of an element
  - **padding-left -** sets the left padding of an element
- All the padding properties can have the following values:
  - **length -** specifies a padding in px, pt, cm, etc.
  - **% -** specifies a padding in % of the width of the containing element
  - **inherit -** specifies that the padding should be inherited from the parent element

Padding - Shorthand Property

- The padding property is a shorthand property for the following individual padding properties:
  - padding-top
  - padding-right
  - padding-bottom
  - padding-left
- Here is how it works:
- Example:

```
div {
  padding: 25px 50px 75px 100px;
}
```

CSS Font

- CSS font properties define the font family, boldness, size, and the style of a text
- CSS Font Families
  - two types of font family names
    - **generic family** - a group of font families with a similar look (like "Serif" or "Monospace")
    - **font family** - a specific font family (like "Times New Roman" or "Arial")

Font Family

- The font family of a text is set with the font-family property.
- The font-family property should hold several font names as a "fallback" system.
- If the browser does not support the first font, it tries the next font.
- Start with the font you want, and end with a generic family, to let the browser pick a similar font in the generic family, if no other fonts are available.
- If the name of a font family is more than one word, it must be in quotation marks, like font-family: "Times New Roman".

Example

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>Setting font-family in CSS</title>
<style>
    body {
        font-family: Arial, Helvetica, sans-serif;
    }
</style>
</head>
<body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
</body>
</html>
```

Font Style

- The font-style property is mostly used to specify italic text.
- This property has three values:
  - normal - The text is shown normally
  - italic - The text is shown in italics
  - oblique - The text is "leaning" (oblique is very similar to italic, but less supported)
- Example
  - p.normal {font-style:normal;}
  - p.italic {font-style:italic;}

Font Size

- The font-size property sets the size of the text.
- The font-size value can be an absolute, or relative size
- **Absolute size:**
  - Sets the text to a specified size
  - Does not allow a user to change the text size in all browsers (bad for accessibility reasons)
  - Absolute size is useful when the physical size of the output is known
- **Relative size:**
  - Sets the size relative to surrounding elements
  - Allows a user to change the text size in browsers
- If you do not specify a font size, the default size for normal text, like paragraphs, is 16px (16px=1em).

Set Font Size with Pixels

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- **Relative size:**
  - Sets the size relative to surrounding elements
  - Allows a user to change the text size in browsers
- If you do not specify a font size, the default size for normal text, like paragraphs, is 16px (16px=1em).

Set Font Size with Pixels

```
h1 {font-size:40px;}
h2 {font-size:30px;}
p {font-size:14px;}
```

- The example above allows Firefox, Chrome, and Safari to resize the text, but not Internet Explorer.

Set Font Size With Em

- To avoid the resizing problem with Internet Explorer, many developers use em instead of pixels.
- The em size unit is recommended by the W3C.
- 1em is equal to the current font size.
- The default text size in browsers is 16px. So, the default size of 1em is 16px.
- Example

```
h1 {font-size:2.5em;} /* 40px/16=2.5em */
h2 {font-size:1.875em;} /* 30px/16=1.875em */
p {font-size:0.875em;} /* 14px/16=0.875em */
```

Font Weight

- If you want to control the weight of your font (its thickness), using font weight is the best way to go about it.
- It is recommended that you only use font-weight in multiples of 100 (e.g. 200, 300, etc) because any less and you probably will not see any difference.
- The values range from 100 (thin)-900 (thick).
- Example

```
p.normal {font-weight:normal;}
p.light {font-weight:lighter;}
p.thick {font-weight:bold;}
p.thicker {font-weight:900;}
```

Font Variant

- CSS Font Variant allows you to convert your font to all small caps.
- Example

p.normal {font-variant:normal;}
p.small {font-variant:small-caps;}

Text Color

- The color property is used to set the color of the text. The color can be specified by:
  - name - a color name, like "red"
  - RGB - an RGB value, like "rgb(255,0,0)"
  - Hex - a hex value, like "#ff0000"
- The default color for a page is defined in the body selector.
- Example

```
body {color:red;}
h1 {color:#00ff00;}
p.ex {color:rgb(0,0,255);}
```

Text Decoration

- The text-decoration property is used to set or remove decorations from text.
- The text-decoration property is mostly used to remove underlines from links for design purposes:

  a {text-decoration:none;}
- text-decoration allows you to add horizontal lines above, below, or through your text.

  h1 {text-decoration:overline;}

  h2 {text-decoration:line-through;}

  h3 {text-decoration:underline;}

Text Indent

- CSS text-indent is a great way to indent your paragraphs without having to use preformatted HTML tags, (<pre>), or inserting spaces manually ( ).
- You may define your indentation with exact values or percentages.
- Example

p { text-indent: 20px; }
h5 { text-indent: 30%; }

Text Alignment

- The text-align property is used to set the horizontal alignment of a text.
- Text can be centered, or aligned to the left or right, or justified.
- When text-align is set to "justify", each line is stretched so that every line has equal width, and the left and right margins are straight.
- Example

```
h1 {text-align:center;}
p.date {text-align:right;}
p.main {text-align:justify;}
```

Text Transformation

- The text-transform property is used to specify uppercase and lowercase letters in a text.
- It can be used to turn everything into uppercase or lowercase letters, or capitalize the first letter of each word.
- Example

```
p.uppercase {text-transform:uppercase;}
p.lowercase {text-transform:lowercase;}
p.capitalize {text-transform:capitalize;}
```

White Space

- The white-space attribute allows you to prevent text from wrapping until you place a break
 into your text.
- Example

p { white-space: nowrap; }

Word spacing

- With the CSS attribute word-spacing you are able to specify the exact value of the spacing between your words.
- Word-spacing should be defined with exact values
- Example

p { word-spacing: 10px; }

Letter spacing

- With the CSS attribute letter-spacing you are able to specify the exact value of the spacing between your letters.
- Letter-spacing should be defined with exact values.
- Example

p { letter-spacing: 3px; }

Letter spacing

- With the CSS attribute letter-spacing you are able to specify the exact value of the spacing between your letters.
- Letter-spacing should be defined with exact values.
- Example

p { letter-spacing: 3px; }

CSS Links

- HTML links can be styled with many CSS properties, like color, text-decoration, background-color, font-size, font-weight, font-family, etc.).
- Example

```
a {
  color: hotpink;
  background-color: yellow;
  font-weight: bold;
}
```

Styling Links with CSS

- A link has four different states — link, visited, active and hover. These four states of a link can be styled differently through using the following anchor pseudo-class selectors.
  - **a:link:** define styles for normal or unvisited links.
  - **a:visited:**define styles for links that the user has already visited.
  - **a:hover:** define styles for a link when the user place the mouse pointer over it.
  - **a:active:** define styles for links when they are being clicked.

Example

```
a:link {    /* unvisited link */
    color: #ff0000;
    text-decoration: none;
    border-bottom: 1px solid;
}
a:visited {    /* visited link */
    color: #ff00ff;
}
a:hover {    /* mouse over link */
    color: #00ff00;
    border-bottom: none;
}
a:active {    /* active link */
    color: #00ffff;
}
```

Modifying Standard Link Styles

- In all major web browsers such as Chrome, Firefox, Safari, etc. links on the web pages have underlines and uses the browser's default link colors, if you do not set the styles exclusively for them.
- By default, text links will appear as follow in most of the browsers:
    - An unvisited link as underlined blue text.
    - A visited link as underlined purple text.
    - An active link as underlined red text.

Setting Custom Color of Links

- Simply use the CSS **color** property to define the color of your choice for different state of a link. But when choosing colors make sure that user can clearly differentiate between normal text and links.

```
a:link {
    color: #1ebba3;
}
a:visited {
    color: #ff00f4;
}
a:hover {
    color: #a766ff;
}
a:active {
    color: #ff9800;
}
```

Removing the Default Underline from Links

- If you don't like the default underline on links, you can simply use the CSS **text-decoration** property to get rid of it. Alternatively, you can apply other styling on links like background color, bottom border, bold font, etc. to make it stand out from the normal text a little better.

```
a:link, a:visited {
    text-decoration: none;
}
a:hover, a:active {
    text-decoration: underline;
}
```

Making Text Links Look Like Buttons

- You can also make your ordinary text links look like button using CSS. To do this we need to utilize few more CSS properties such as background-color, border, display, padding, etc.

```
a:link, a:visited {
    color: white;
    background-color: #1ebba3;
    display: inline-block;
    padding: 10px 20px;
    border: 2px solid #099983;
    text-decoration: none;
    text-align: center;
    font: 14px Arial, sans-serif;
}
a:hover, a:active {
    background-color: #9c6ae1;
    border-color: #7443b6;
}
```

Thank You!